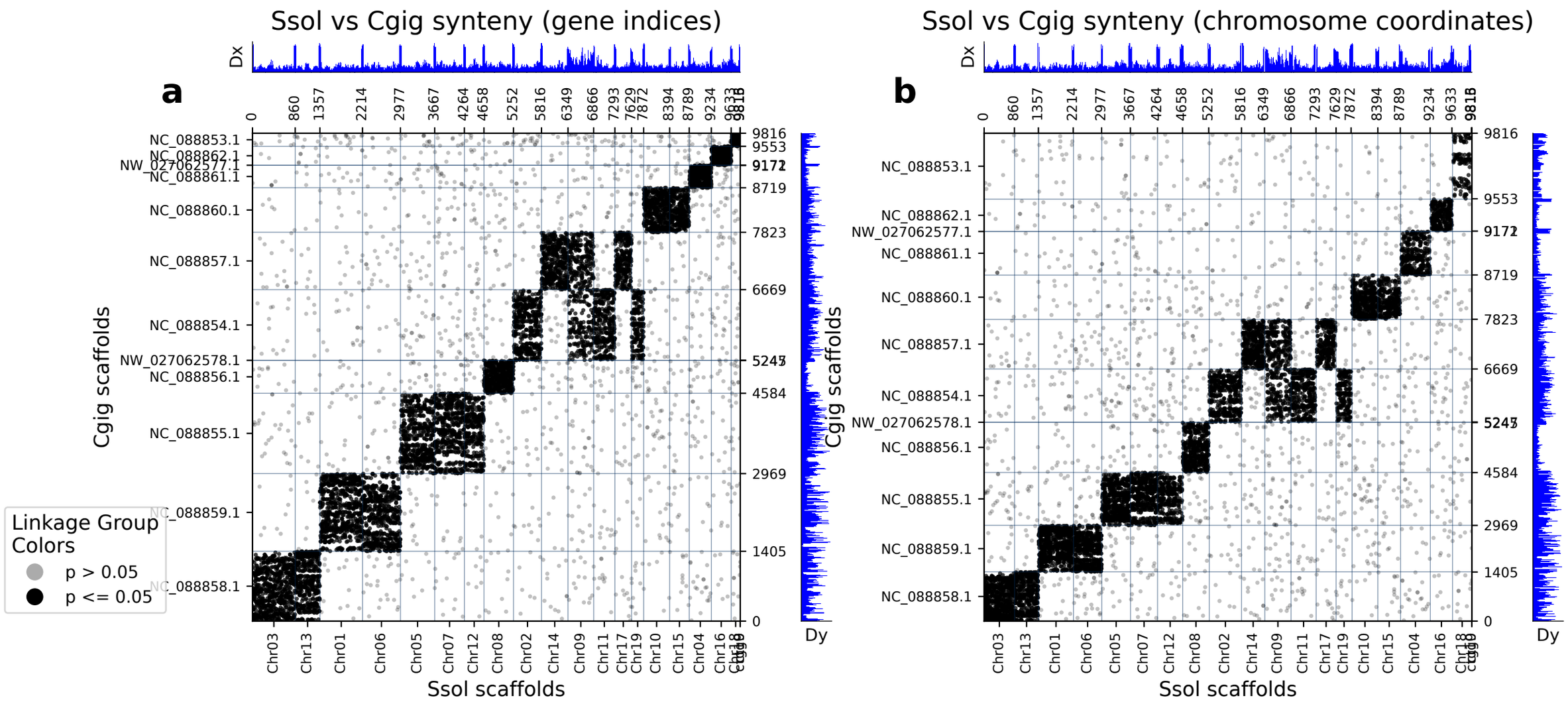
diamond: 9817 orthologs on 21 Ssol scaffolds and 12 Cgig scaffolds.



c

Ssol vs Cgig groupings

d

Ssol vs Cgig FET, whole chroms.

	Ssol scaf	Cgig scaf	p (FET)	Count
0	Chr03	NC_088858.1	0.00e+00	768
1	Chr01	NC_088859.1	0.00e+00	755
2	Chr06	NC_088859.1	0.00e+00	665
3	Chr05	NC_088855.1	0.00e+00	582
4	Chr08	NC_088856.1	0.00e+00	501
5	Chr07	NC_088855.1	1.66e-295	488
6	Chr02	NC_088854.1	0.00e+00	486
7	Chr14	NC_088857.1	0.00e+00	458
8	Chr13	NC_088858.1	0.00e+00	446
9	Chr10	NC_088860.1	0.00e+00	431
10	Chr11	NC_088854.1	6.86e-237	356
11	Chr04	NC_088861.1	0.00e+00	345
12	Chr15	NC_088860.1	4.53e-288	326
13	Chr12	NC_088855.1	2.24e-186	321
14	Chr16	NC_088862.1	0.00e+00	309
15	Chr17	NC_088857.1	4.13e-209	279
16	Chr09	NC_088857.1	6.91e-96	246
17	Chr19	NC_088854.1	2.59e-117	194
18	Chr09	NC_088854.1	1.54e-30	180
19	Chr18	NC_088853.1	6.88e-157	122

e

Ssol vs Cgig FET, chrom. pieces

	Ssol scaf	Cgig scaf	p (FET)	Count
0	Chr03 : 0-end	NC_088858.1 : 0-end	0.00e+00	768
1	Chr01 : 0-end	NC_088859.1 : 0-end	0.00e+00	755
2	Chr06 : 0-end	NC_088859.1 : 0-end	0.00e+00	665
3	Chr05 : 0-end	NC_088855.1 : 0-end	0.00e+00	582
4	Chr08 : 0-end	NC_088856.1 : 0-end	0.00e+00	501
5	Chr07 : 0-end	NC_088855.1 : 0-end	1.66e-295	488
6	Chr02 : 0-end	NC_088854.1 : 0-end	0.00e+00	486
7	Chr14 : 0-end	NC_088857.1 : 0-end	0.00e+00	458
8	Chr13 : 0-end	NC_088858.1 : 0-end	0.00e+00	446
9	Chr10 : 0-end	NC_088860.1 : 0-end	0.00e+00	431
10	Chr11 : 0-end	NC_088854.1 : 0-end	6.86e-237	356
11	Chr04 : 0-end	NC_088861.1 : 0-end	0.00e+00	345
12	Chr15 : 0-end	NC_088860.1 : 0-end	4.53e-288	326
13	Chr12 : 0-end	NC_088855.1 : 0-end	2.24e-186	321
14	Chr16 : 0-end	NC_088862.1 : 0-end	0.00e+00	309
15	Chr17 : 0-end	NC_088857.1 : 0-end	4.13e-209	279
16	Chr09 : 0-end	NC_088857.1 : 0-end	6.91e-96	246
17	Chr19 : 0-end	NC_088854.1 : 0-end	2.59e-117	194
18	Chr09 : 0-end	NC_088854.1 : 0-end	1.54e-30	180
19	Chr18 : 0-end	NC_088853.1 : 0-end	6.88e-157	122

(a) depicts the chromosomes/scaffolds of Ssol (x-axis) plotted against the chromosomes/scaffolds of Cgig (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 9817 orthologs found between 21 Ssol scaffolds and 12 Cgig scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.